

Generation tokens — pricing

AI-MODEL-CHAT · GENERATION STUDIO · DECISION DOCUMENT

The Generation Studio billing unit, as shipped. **Option B, raised allowances and no tier discount bands** were chosen; sections 4 and 9 keep the options and the reasoning so a settled question is not re-opened.

Implemented in `credits.py`, `db.py`, `app.py` and `studio.html`; the numbers below are the ones the code asserts at import. Generation itself is still admin-only — this prices the feature, it does not open it.

1. What is wrong with credits

A photo costs **20 credits**. A five-second clip costs **230**. Starter includes **600 a month**, Pro 2,500, Agency 7,500.

None of those numbers mean anything to a creator. The scale exists because `CREDIT_COST_USD = 0.002` made one credit a fixed slice of *provider cost* — excellent for margin discipline, useless as a thing a person reads before pressing Generate. A creator cannot tell whether 230 is a lot.

Two changes:

1. **Re-denominate to tokens at human scale.** One token is about one photo. A clip is about twelve. A month's allowance is a two- or three-digit number.
2. **Cut the retail price from ~15x provider cost to 2.3–3.8x**, on a ladder that discounts 40% by volume. A photo is **€0.15** at the smallest pack and **€0.09** at the largest — at or below every comparable platform.

The margin discipline is kept exactly: a token is still a fixed slice of provider cost, the slice is just twenty times bigger.

2. What the market charges

The comparison set is deliberately two: **Higgsfield**, the retail product a creator would otherwise buy, and **WaveSpeed**, an inference provider we could buy from instead of Runware. One is a competitor for the customer, the other a competitor for our cost base.

	photo	5s 720p clip	NSFW	our cost
Us	€0.15 → €0.09	€1.80 → €1.08	yes, identity-locked	\$0.040 / \$0.504
Higgsfield, \$19 tier	\$0.141	\$0.56	no	—
Higgsfield, \$99 tier	\$0.066	\$0.26	no	—
WaveSpeed (a provider)	\$0.040 Seedream	\$1.80 Seedance 720p	n/a	it is cost

Our range is smallest pack to largest. Higgsfield's is entry tier to top tier — its credits get cheaper the more you subscribe, the same shape as our pack ladder.

What €130 buys

	tokens or credits	photos	5s clips
Us (1,000-token pack)	1,000	1,000	83
Higgsfield at its \$19 rate	~1,995	998	249
Higgsfield at its \$99 rate	~4,255	2,127	532

On photos we are level with Higgsfield's entry tier and behind its top tier. On video we are 3–6x behind, and this is a cost problem, not a margin one. Runware bills us \$0.09076 a second for Wan 2.5; fal.ai lists \$0.05 and EvoLink \$0.0708. Higgsfield's top tier sells a 5s Wan 2.7 clip for **\$0.264 — less than the \$0.504 it costs us**. No pricing decision closes that gap; a cheaper provider might. See section 8.

WaveSpeed is in the table as the provider alternative and does not beat Runware where it matters: Seedream is the same \$0.04, and its Seedance 2.5 at 720p is \$1.80 per 5s against Runware's \$0.60. Its cheap rung is *Wan 2.2 Ultra Fast* at \$0.01 a second, which is a different, lower-quality model rather than the same clip for less.

The one thing neither of them sells

Higgsfield refuses NSFW outright, and WaveSpeed is an API you would have to build this on top of. **Nothing in the comparison set generates explicit content of a specific creator's face**, conditioned on an approved vault photo and faceswapped from the same photo. That is the product; the per-photo price is not what a creator is choosing between.

3. What a token costs us

Unchanged from today, just re-pegged. `TOKEN_COST_USD = 0.04` — what one token is allowed to cost us at the provider. Every generation is priced `ceil(provider_cost / TOKEN_COST_USD)`, so rounding always favours us.

Measured provider costs (from live bills, in `credits.PROVIDER_COST_USD`):

	cost
Seedream 4.5 / 5.0 Pro, any size	\$0.04
Nano Banana 2, 2k	\$0.10255
Nano Banana Pro, 2k	\$0.138
Wan 2.5, per second 720p	\$0.09076
Wan 2.7, per second 720p	\$0.10076

Everything else in the table is a deliberate over-estimate. **That mattered little at 15x margin and matters a great deal at 2.3–3.8x** — see section 8.

4. The three options

Option A — flat rate

One token per photo, ten per clip, whatever model you pick. The clearest pricing a person could be given.

It does not work. At €0.15 a token — the smallest pack's rate — here is what a flat rate actually earns against what it costs:

Generation	Costs us	Flat price	Earns	Multiple	
Photo — Seedream 2k/4k	\$0.040	1 token	\$0.153	3.82x	OK
Photo — Nano Banana 2, 2k	\$0.103	1 token	\$0.153	1.49x	under floor
Photo — Nano Banana Pro, 2k	\$0.138	1 token	\$0.153	1.11x	under floor
Photo — Nano Banana Pro, 4k	\$0.276	1 token	\$0.153	0.55x	loses money
Clip — Wan 2.5, 5s 720p	\$0.454	10 tokens	\$1.530	3.37x	OK
Clip — Wan 2.7, 5s 720p	\$0.504	10 tokens	\$1.530	3.04x	OK
Clip — Wan 2.5, 5s 1080p	\$1.135	10 tokens	\$1.530	1.35x	under floor
Clip — Seedance, 10s 1080p	\$3.000	10 tokens	\$1.530	0.51x	loses money

To clear the 2.25x floor flat we would need a **5-token photo** and a **9-token clip** — at which point it is no longer flat in any useful sense. The only way to keep "1 token = 1 photo" flat is to **remove every image model except Seedream and cap video at 720p**, which throws away Nano Banana Pro and the whole premium tier.

Recorded so it is not re-proposed. At 15x margin this worked; at these prices it cannot.

Option B — graded, 1 token $\approx$ 1 photo (*recommended*)

A standard photo is 1 token. Better models cost more, in single digits.

Images

Model	2k	4k
Seedream 4.5	1	1
Seedream 5.0 Pro	1	1
Nano Banana 2	3	6
Nano Banana Pro	4	7

Video (whole job rounded up, so any length is priced)

Model	720p 3s	5s	10s	1080p 3s	5s	10s
Wan 2.5	7	12	23	18	29	57
Wan 2.7	8	13	26	19	32	63
Seedance 2.5	9	15	30	23	38	75
Face swap	9	15	30	23	38	75

Audio add-on: **3 tokens** a clip. Upscale and the NSFW check disappear as separate line items — at this scale they round to 1 token, which would be a 10x overcharge on a \$0.004 operation, so they fold into the base price.

Monthly allowance, as shipped: **100 / 350 / 1,000** (Starter / Pro / Agency). Demo keeps unlimited generation, so it has no allowance to state.

Option C — graded, 2 tokens = 1 photo

Identical shape, doubled. Matches the Candy AI and Higgsfield convention exactly, so a creator arriving from either reads our prices without conversion.

Model	2k	4k	Model	720p 5s	1080p 5s
Seedream 4.5	2	2	Wan 2.5	23	57
Nano Banana 2	6	11	Wan 2.7	26	63
Nano Banana Pro	7	14	Seedance 2.5	30	75

Audio: 5 tokens. Allowance would have been **200 / 700 / 2,000**.

The case for C is finer grading — the gap between Seedream and Nano Banana Pro is 2 → 7 rather than 1 → 4, so a price difference is visible without being dramatic. The case against is that every number is twice as big for no extra information, and "1 token = 1 photo" is the single sentence that makes the whole system explainable.

Side by side

	A — flat	B — 1 tk/photo	C — 2 tk/photo
Standard photo	1	1	2
Premium photo 4k	1	7	14
5s 720p clip	10	12	23
Pro allowance	350	350	700
Explainable in one line	yes	yes	nearly
Clears the 2.25x margin floor	no	yes	yes
Needs models removed	yes	no	no

5. What a token costs, in euro, dollars and pounds

Euro is the base currency. Prices are set in euro — matching the existing €49 / €149 / €349 plans — and the dollar and pound ladders are hand-set round numbers alongside, not live conversions. A pack never costs €13.47 and never moves because the exchange rate did.

The floor is still checked in dollars, because the providers bill us in dollars. Each euro and pound price is converted at a **conservative** reference rate (€1 = \$1.02, £1 = \$1.18) and must still clear 3.0x. Using a pessimistic rate means an ordinary FX swing cannot quietly push a price under cost.

The ladder — one price for everyone

Tokens	EUR	USD	GBP	€/token	margin	buys
100	€15	\$16	£13	€0.150	3.82x	100 photos / 8 clips
500	€70	\$75	£62	€0.140	3.57x	500 photos / 41 clips
1,000	€130	\$139	£115	€0.130	3.31x	1,000 photos / 83 clips
2,000	€220	\$235	£195	€0.110	2.81x	2,000 photos / 166 clips
5,000	€450	\$479	£395	€0.090	2.29x	5,000 photos / 416 clips

A photo runs **€0.15 down to €0.09** and a five-second clip **€1.80 down to €1.08**, depending on pack size. Generation prices quoted in cash use the smallest pack's rate, because that is the marginal price of buying more — the same reason `credits.credit_rate_usd()` reads `PACK_SIZES[0]` today.

MIN_MARGIN_MULTIPLE moves **4.0 → 2.25**. The two largest packs sell at 2.81x and 2.29x, under the 3.0x the smaller ones clear. That is the volume discount working as intended — gross margin is still 64% and 56% — but the floor is a build-gate assertion that runs at import, so it must come down or nothing starts.

The ladder already is the discount — drop the tier bands

Today Pro and Agency buy credits ~13% and ~27% cheaper than Starter. This ladder already falls 40% from smallest pack to largest, which is the same incentive bought by volume rather than by subscription tier. Stacking the tier bands on top would put the 5,000 pack at roughly **1.7x cost**.

Recommendation: drop the bands. Price on size alone, and let the upgrade incentive live in the included allowance instead — which is what section 6 is about.

6. The allowance, raised

Re-denominating exposed something the old scale was hiding. What each plan used to include, read in clips rather than credits:

Plan	Price	Was	= Photos	= Clips	Cost us
Starter	€49	600 credits	30	2	\$1.20
Pro	€149	2,500 credits	125	10	\$5.00
Agency	€349	7,500 credits	375	31	\$15.00

Starter was €49 a month for two video clips. For a product whose entire pitch is "generate content your fans will pay for" that is not a credible entry tier — and at the old €0.56-per-photo pricing it at least *sounded* substantial.

Shipped instead, roughly 3x across the board:

Plan	Now	= Photos	= Clips	Costs us/mo	% of plan price
Starter	100	100	8	\$4.00	8%
Pro	350	350	29	\$14.00	9%
Agency	1,000	1,000	83	\$40.00	11%

Single-digit percentages of revenue, and it restores the upgrade ladder that dropping the pack discount bands gives up. The numbers live once, in `credits.MONTHLY_TOKENS`; the tier bullets and the enforced capability both read them from there, so a marketing string cannot drift from what the plan allows — `test_tokens.py` asserts it.

7. What is not changing

- **Monthly allowance still expires monthly.** No rollover. Bought tokens never expire. Same as Kling and Higgsfield, same as the code does today.
- **Generation stays admin-only.** `/studio` and every `/api/generate/*` route keep `_require_admin` and keep 404-ing rather than 403-ing. Token checkout stays shut too — nobody should buy tokens for a feature that is not offered yet. This document prices the feature; it does not open it.
- **The ledger stays append-only.** Balance is the sum of rows, never a counter, because people will buy these. A spend still drains the expiring allowance before anything purchased; a refund still returns tokens to the bucket they left.
- **Pricing stays pegged to provider cost.** One number, `TOKEN_COST_USD`. A new model is a table entry, not a pricing decision.

8. The risk this creates

Most of the price table is not measured. From `credits.py`'s own notes, the following are deliberate over-estimates carried because *a guess that is too low loses money on every generation and nothing reports it*:

- **Seedance 2.5** — refused the probe at ByteDance's moderation end before billing anything. Entire row is a guess.
- **Every 4k image rung** — priced at twice 2k because the 4k result came back before the provider reported a cost.
- **480p and 1080p video** — only 720p was measured. 480p is charged at the 720p rate; 1080p keeps the old table's 2.5x shape.
- **The swap models** (`p-video-replace`, `ml-face-swap`) — unmeasured.

- **The audio add-on** — \$0.10 assumed, and per `CLAUDE.md` none of the audio model ids or field names have been verified against the live catalogue.

At 15x margin a 2x over-estimate was invisible. **At 2.3x on the largest pack it is the difference between a healthy margin and selling under cost.** Before this ships, measure at minimum: Seedance at 720p, one 4k image on each Google model, and one 1080p clip. Run `imagegen.search_models('audio')` against a live Runware key and correct the audio defaults.

The build gate protects us either way — `_assert_generation_floor()` walks the entire job matrix at import and refuses to start if any rung would sell under cost — but it can only check the numbers it is given.

9. Decisions — settled

1. **Which denomination?** → **B.** One token, one photo.
2. **Raise the included allowances?** → **yes:** 100 / 350 / 1,000 for Starter / Pro / Agency. Demo keeps unlimited generation, so it has no allowance to state.
3. **Drop the tier discount bands?** → **yes.** The size ladder already discounts 40%.

All three are implemented. What remains open is the video cost gap in section 2: measure Wan on fal.ai and WaveSpeed against Runware's \$0.09076 a second, and if a cheaper route holds, clips fall below 12 tokens at the same margin.

Once these are settled the implementation is mechanical: `credits.py` and its tests first (pure, no I/O, the floor assertions prove the numbers before anything else moves), then the ledger, then the routes and checkout, then the studio's own labels.